8-4. Model checking: Analysis of deviance and residuals for GLMs

- To fit a GLM to data (that is, estimate the regression parameters), we use Maximum Likelihood.
- That is, maximise the log-likelihood $l(\boldsymbol{\beta})$
- To assess goodness-of-fit, compare model of interest with the 'saturated model', which has one parameter for each observation.
- If $l(\text{fitted model})$ is close to $l(\text{saturated model})$, then the model is a good fit.

- We can assess the fit of the model using the **deviance**.
- The deviance of the model is given by

$$D = -2\,(\,l(\text{fitted model}) - l(\text{saturated model})\,)$$

where

  - $l(\text{fitted})$ is the **log-likelihood evaluated at the estimated parameter(s)** (e.g. at the MLE $\hat{\boldsymbol{\beta}}$)
  - the **saturated** (or full, or maximal) model is the **model with a different parameter for each observation**

- D is equal to $-2 \times \log(\text{likelihood ratio statistic})$
- The deviance can be used to check the fit of a model, and also to compare the fit of two nested models.

Example: Deviance for logistic regression

Consider the model

$$D_i \sim \text{Bin}(E_i, q_i), \quad \text{independent.}$$

Likelihood function is

$$L(\boldsymbol{q}; \boldsymbol{d}) = \prod_{i=1}^{n} \binom{E_i}{d_i} q_i^{d_i} (1-q_i)^{E_i - d_i}$$

so log-likelihood is

$$l(\boldsymbol{q}; \boldsymbol{d}) = \sum_{i=1}^{n} \left\{ \log \binom{E_i}{d_i} + d_i \log q_i + (E_i - d_i) \log(1-q_i) \right\}$$

To find MLEs, differentiate w.r.t. each q_k:

$$\frac{\partial l}{\partial q_k} = \frac{d_k}{q_k} - \frac{E_k - d_k}{1 - q_k} \text{ for each } k$$

At a maximum point, $\frac{\partial l}{\partial q_k} = 0$ for $k = 1, 2, \ldots, n$, and so we find

$$\hat{q}_k = \frac{d_k}{E_k} \text{ for } k = 1, 2, \ldots, n.$$

(Natural estimate — use observed proportion to estimate death probability.)

Note: So far, we have not introduced any link function. This is the *maximal model*, or *saturated model* — a separate parameter q_k for each observation, with no assumed relationship between them.

Now suppose we measure some explanatory variable x_i, and assume the logistic link

$$\log\left(\frac{q_i}{1-q_i}\right) = \beta_0 + \beta_1 x_i$$

Log-likelihood is

$$\begin{aligned} l(\boldsymbol{q};\boldsymbol{d}) &= \sum_{i=1}^{n}\left\{\log\binom{E_i}{d_i} + d_i \log q_i + (E_i - d_i)\log(1-q_i)\right\} \\ &= const + \sum_{i=1}^{n}\left\{d_i \log\left(\frac{q_i}{1-q_i}\right) + E_i \log(1-q_i)\right\} \end{aligned}$$

With logistic link,

$$l\left(\beta_0, \beta_1; \boldsymbol{d}\right) \quad = \quad const + \sum_{i=1}^{n} \left\{ d_i \left(\beta_0 + \beta_1 x_i\right) - E_i \log\left(1 + \mathrm{e}^{\beta_0 + \beta_1 x_i}\right) \right\}$$

Differentiate w.r.t. each parameter:

$$\begin{aligned} \frac{\partial l}{\partial \beta_0} &= \sum_{i=1}^{n} \left\{ d_i - \frac{E_i \, \mathrm{e}^{\beta_0 + \beta_1 x_i}}{1 + \mathrm{e}^{\beta_0 + \beta_1 x_i}} \right\} \\ \frac{\partial l}{\partial \beta_1} &= \sum_{i=1}^{n} \left\{ d_i x_i - \frac{E_i \, \mathrm{e}^{\beta_0 + \beta_1 x_i} x_i}{1 + \mathrm{e}^{\beta_0 + \beta_1 x_i}} \right\} \end{aligned}$$

To find MLEs, solve $\frac{\partial l}{\partial \beta_0} = \frac{\partial l}{\partial \beta_1} = 0$ for $\hat{\beta}_0, \hat{\beta}_1$. Needs to be done numerically, in R.

Recall definition of deviance:

$$D \;=\; -2\,(\,l(\text{fitted model}) - l(\text{saturated model})\,)$$

For maximal model, have MLEs $\hat{q}_i = d_i/E_i$, and

$$l\,(\hat{\boldsymbol{q}};\boldsymbol{d}) \;=\; \sum_{i=1}^{n}\left\{\log\binom{E_i}{d_i} + d_i\log\left(\frac{\hat{q}_i}{1-\hat{q}_i}\right) + E_i\log(1-\hat{q}_i)\right\}$$

For fitted model of interest, MLEs $\hat{\beta}_0, \hat{\beta}_1$ must be computed numerically, and then

$$l\left(\hat{\beta}_0,\hat{\beta}_1;\boldsymbol{d}\right) \;=\; \sum_{i=1}^{n}\left\{\log\binom{E_i}{d_i} + d_i\left(\hat{\beta}_0+\hat{\beta}_1 x_i\right) - E_i\log\left(1+\mathrm{e}^{\hat{\beta}_0+\hat{\beta}_1 x_i}\right)\right\}$$

So putting this together, for the logistic regression model we find

$$\begin{aligned} D &= -2\sum_{i=1}^{n}\left\{d_i\left(\hat{\beta}_0+\hat{\beta}_1 x_i\right)-E_i\log\left(1+\mathrm{e}^{\hat{\beta}_0+\hat{\beta}_1 x_i}\right)\right. \\ &\qquad \left. -d_i\log\left(\frac{d_i}{E_i-d_i}\right)-E_i\log\left(1-\frac{d_i}{E_i}\right)\right\} \end{aligned}$$

Could try to simplify this further, but not a lot of point — need to use R to numerically compute $\hat{\beta}_0, \hat{\beta}_1$ anyway, and then R will also compute the deviance.

Exercise: find an expression for the deviance for the log-linear trend model with Poisson response. (You can check your answer by evaluating your expression with the data of example 8.1 and comparing with the deviance value computed by R.)

Analysis of deviance

Test the model fit:

H_0: model fit is adequate vs. H_1: fit not adequate

- Under the hypothesis that model fit is good (H_0), it can be shown that

$$D \sim \chi^2_{n-p} \qquad \text{(approximately)}$$

  where n is the number of observations and p the number of parameters fitted.

- **Good models have small deviance.**
  Therefore, reject H_0 for large D (one-sided test).
  Use critical values from χ^2_{n-p}, or use R to compute p-value.

Example 8.1 (Poisson log-linear model)

Recall the data of example 8.1: numbers of stressful events reported by a sample of people for the previous 18 months.

```
> stressmodel=glm(n~x, family=poisson)
> summary(stressmodel)

Call:
glm(formula = n ~ x, family = poisson)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-1.9886  -0.9631   0.1737   0.5131   2.0362

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  2.80316    0.14816  18.920  < 2e-16 ***
x           -0.08377    0.01680  -4.986 6.15e-07 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 50.843  on 17  degrees of freedom
Residual deviance: 24.570  on 16  degrees of freedom
AIC: 95.825
```

- From `summary` output, Residual deviance $= 24.570$. Have 18 observations, fitting a model with $p = 2$ regression parameters, so df $= 18 - 2 = 16$.
- To test goodness-of-fit, compare deviance with χ^2_{16} distribution, so p-value is `1 - pchisq(24.570, df=16)` = `0.07777325`. Reject goodness-of-fit at the 10% level, but not at the 5% level.
- That is, at the 5% level, the fitted log-linear model is not a significantly worse fit than the maximal model.
- This model is a reasonably good fit to the data ($p = 0.078$ for deviance goodness-of-fit test).

Example 8.3 (Binomial logistic model)

Recall the data of example 8.3: 10 groups of insects exposed to different doses of a drug.

```
> summary(drugmodel)
Call:
glm(formula = propdead ~ dose, family = binomial, weights = groupsize)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-1.8635  -0.5465   0.2424   1.0777   2.7251

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -20.2724     1.5956  -12.71   <2e-16 ***
dose          4.7331     0.3661   12.93   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 298.067  on 9  degrees of freedom
Residual deviance:  17.234  on 8  degrees of freedom
AIC: 57.3
```

- From `summary` output, Residual deviance $= 17.234$. Have 10 observations, fitting a model with $p = 2$ regression parameters, so df $= 10 - 2 = 8$.
- For testing goodness-of-fit, p-value is `1 - pchisq(17.234, df=8) = 0.02776262`. Reject goodness-of-fit at the 5% level (although not at the 1% level).
- That is, at the 5% level, the fitted logistic regression model is a significantly worse fit than the maximal model.
- Model is not a very good fit to the data ($p = 0.028$ for deviance goodness-of-fit test).
- Recall we previously saw that a plot of residuals for this fitted model showed a clear pattern. Both the residuals plot and the deviance goodness-of-fit test suggest that this model can be improved.

Model residuals

Let $\mu_i = \mathsf{E}[Y_i|X_i]$, and $\hat{\mu}_i$ the corresponding fitted value.

Ordinary ('`response`') residuals $(y_i - \hat{\mu}_i)$ are not generally used in GLMs (as they have non-constant variance). Instead use:

- **Pearson residuals**

$$r_i^P = \frac{y_i - \hat{\mu}_i}{\sqrt{\mathsf{var}(\hat{\mu}_i)}}$$

- **Deviance residuals**

$$r_i^D = \mathsf{sign}(y_i - \hat{\mu}_i)\sqrt{d_i}$$

where d_i is the term of the deviance, D, associated with y_i

To understand the definition of deviance residuals, note that the log-likelihood for the whole sample is the sum of the log-likelihoods of individual observations,

$$l(\boldsymbol{\beta}; \boldsymbol{y}) \quad = \quad \sum_{i=1}^{n} l\left(\boldsymbol{\beta}; y_i\right)$$

and the deviance may be written as

$$D \quad = \quad \sum_{i=1}^{n} \left\{ -2 \left(l\left(\widehat{\boldsymbol{\beta}}; y_i\right) - l\left(\widehat{\boldsymbol{\mu}}; y_i\right) \right) \right\} \quad = \quad \sum_{i=1}^{n} d_i$$

Recall Multiple Linear Regression: estimate parameters by minimising Sum of Squared residuals

$$SS = \sum_{i=1}^{n} e_i^2$$

By analogy with MLR, write

$$D = \sum_{i=1}^{n} d_i = \sum_{i=1}^{n} \left(r_i^D\right)^2$$

and choose the sign of r_i^D to agree with that of the raw residual $(y_i - \hat{\mu}_i)$

That is, $r_i^D = \text{sign}\,(y_i - \hat{\mu}_i)\sqrt{d_i}$

R gives deviance residuals by default.

Note: for a GLM with non-normal response variable,

- Do **not** expect residuals to be normally distributed, so no point looking at QQ plot of residuals.
- Do **not** expect homoscedasticity. For instance, with Poisson response variable, variance increases as mean response μ_i increases (variance $=$ mean for Poisson distribution).